

Taeyun Roh

<https://txxnrd.github.io>

Education

- **Korea University**, Seoul, Republic of Korea *Feb. 2026 – Present*
 - Master of Science in Computer Science and Engineering
- **Korea University**, Seoul, Republic of Korea *Mar. 2022 – Feb. 2026*
 - Bachelor of Science in Computer Science and Engineering

Research Experience

- **Data Mining Information Systems Laboratory, Korea University**
Advisor: Professor Jae Woo Kang *Research Intern and MS Student, March. 2025 –*
 - Research on Agentic Memory system for AI agents
- **Network and Mobile System Laboratory, KAIST**
Advisor: Professor Sung-Ju Lee *Research Intern, Jan. 2025 – March.2025*
 - Focused on Efficient LLM inference including on-device LLM inference
 - Developed LLM-based methods to detect the malicious Spymic app, enhancing threat identification accuracy.
- **System Software Laboratory, Korea University**
Advisor: Professor Gyeongsik Yang *Research Intern, Jan. 2024 – Dec.2024*
 - Research about Federated Learning in Healthcare Domain in aware of Resource heterogeneity

Development & Related Experience

- **Aigensciences** *AI Researcher, March 2025 – August 2025*
 - Researching LLM agent for Biomedical Domain.
 - Machine Learning algorithms for Drug Discovery
- **Blurting** *Frontend Developer, June 2023 – Present*
 - Developed a blind dating app with a completely new format. App Store link
 - Ranked 73rd in App Store Social Networking Category in Korea.
- **Hiconsy** *Data Analyst, Jan. 2022 – Jan. 2023*
 - Worked as a data analyst in education startup Hiconsy, now one of the biggest education companies in Korea.
 - Conducted advanced mathematical modeling to optimize the Teaching Assistant (TA) compensation system and performed regression analysis to investigate the relationship between student study time and academic performance

Awards

- **1st Place in Korea University Capstone Design (₩1,800,000 Prize)**
 - Developed *NEWSQRAB*, an AI-powered automated short-form video generation service for news content using Retrieval-Augmented Generation (RAG)-based LLMs.
- **2nd Place at Korea University NEXT Software Competition(₩1,000,000 Prize)**
 - Participated as an AI Engineer in the project *NEWSQRAB*, an automated short video creation service using AI.
- **3rd Place Award in the Undergraduate Category at the 2024 KIISE Conference**
 - Awarded for the paper '*Analysis on Research for Data and Resource Heterogeneity in Split-Federated Learning*'.

2nd Place in 2023 Korea University Hackathon(₩1,000,000 Prize)

- Developed a mobile application named *Noonchi* that predicts the number of visitors in Han River parks, providing real-time recommendations for less crowded parks based on user preferences.
- Check out the project on GitHub.

Extracurricular Activities

- **DevKor** *Regular Member, June 2023 – Present*
 - Contributed to the frontend of the Blurting Project using Flutter.
 - Enhanced knowledge of frontend technologies and cross-platform development.
 - For more information, visit DevKor's page.
- **FC COMING** *Founder and Captain, June 2023 – Present*
 - Founder of a soccer club in Korea University College of Informatics.
 - Participated in Korea University Sports Soccer League (KU LEAGUE) in 2023 and 2024.

Projects

- **Xalute: Wearable Health Monitoring App with Korea University Anam Hospital**
 - Collaborated with Korea University Anam Hospital to develop a digital healthcare mobile application that interfaces with wearable devices, enabling real-time health tracking and intelligent alerts.
- **Kakao Pediatric Response Chatbot**
 - Co-developed a pediatric Q&A chatbot system in collaboration with Kakao Healthcare and Upstage, enabling parents to receive real-time responses for child healthcare concerns using large language models.

- **K-Mellody: Federated Learning-Based Model for Pharmaceutical Collaboration**

- Participated in the development of *K-Mellody*, a federated learning framework enabling secure and privacy-preserving collaboration across pharmaceutical companies.
- Conducted model architecture research and training pipeline integration at Aigen Sciences.

Technical Skills

- **Frontend:** HTML, CSS, React, Flutter
- **Programming Languages:** Python, C, C++, JavaScript, Dart, Rust

Languages

- **English:** TOEIC : 940